5.2.2 LU factorization: The right-looking algorithm

Bookmark this page

Week 05 due Feb 12, 2025 20:19 IST

Video

other, the corresponding

parts. So what we conclude is that alpha_11 must be equal to upsilon_11, a_21,

sorry, a_12 transpose must be equal to u_12 transpose, a 21 must be equal to l_21

times upsilon_11, which, by the way, we could have written as upsilon11 times

l_21 because a vector times the scalar is the same as a

scalar times a vector. And finally, A_22 is equal to -- let's see-- l_21 times u_12

8:30 / 8:30

2.0x

Video

Download video file

Transcripts

Download SubRip (.srt) file

Download Text (.txt) file

Reading assignment

1/1 point (graded)

Read Unit 5.2.2 of the notes. [LINK]

- [x] done

Submit

Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 05 / 05.2.2

Add a Post

Show all posts | by recent activity

There are no posts in this topic yet.

Homework 5.2.2.1

1/1 point (graded)

Choose the approximate cost incurred by

$A = \text{LU-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right)$

A_{TL} is 0×0

while $n(A_{TL}) < n(A)$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$

$a_{21} := a_{21}/\alpha_{11}$

$A_{22} := A_{22} - a_{21}a_{12}^T$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$

endwhile

when applied to an $n \times n$ matrix.

- ( ) $2n^3$
- (•) $\frac{2}{3}n^3$
- ( ) $\frac{1}{3}n^3$

Solution:

For a complete solution, see the notes .

Submit

Answers are displayed within the problem

Homework 5.2.2.2

1/1 point (graded)

Choose the approximate cost incurred by

$A = \text{LU-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right)$

A_{TL} is 0×0

while $n(A_{TL}) < n(A)$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$

$a_{21} := a_{21}/\alpha_{11}$

$A_{22} := A_{22} - a_{21}a_{12}^T$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$

endwhile

when applied to an $m \times n$ matrix.

- (•) $mn^2 - \frac{1}{3}n^3$
- ( ) $\frac{2}{3}n^3$
- ( ) $\frac{1}{2}mn^2 - \frac{1}{6}n^3$

Submit

✔ Correct (1/1 point)

Homework 5.2.2.3

1/1 point (graded)

Implement the algorithm

$A = \text{LU-right-looking}(A)$

$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$

A_{TL} is 0×0

while $n(A_{TL}) < n(A)$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$

$$a_{21} := a_{21} / \alpha_{11}$$

$$A_{22} := A_{22} - a_{21} a_{12}^T$$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array} \right)$$

endwhile

as

```
function [ A_out ] = LU_right_looking( A )
```

by completing the code in Assignments/Week05/matlab/LU_right_looking.m. Input is an m \times n matrix A. Output is the matrix A that has been overwritten by the LU factorization. You may want to use Assignments/Week05/matlab/test_LU_right_looking.m to check your implementation.

☑ done

✔

Submit

✔ Correct (1/1 point)

‹ Previous Next ›

edX

[About](#)
[Affiliates](#)
[edX for Business](#)
[Open edX](#)
[Careers](#)
[News](#)

Legal

[Terms of Service & Honor Code](#)
[Privacy Policy](#)
[Accessibility Policy](#)
[Trademark Policy](#)
[Sitemap](#)
[Cookie Policy](#)
Your Privacy Choices

Connect

[Idea Hub](#)
[Contact Us](#)
[Help Center](#)
[Security](#)
[Media Kit](#)

GET IT ON Google play
Download on the App Store